

AttenDID: Attendance System Using Virtual Card

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Abstract—Tracking students' attendance is still a hassle in most universities, especially in packed classes. Most of them still rely on the traditional method, like sign-in sheets, which can be cumbersome, liable to cheating, or impractical. The project suggests a smart attendance system using Near Field Communication (NFC) technology to make student attendance easy and digitize it in institutions. The system allows students to take attendance by scanning an NFC tag on an NFC device of a lecturer, thus avoiding physical sign-in procedures and reducing attendance forgery. The system is implemented as a mobile application developed using Flutter, supporting both Android and iOS platforms. Firebase is the backend for user authentication, real-time database operations, and cloud storage, ensuring data synchronization and accessibility. By integrating the NFC technology into a simple-to-use mobile, the system enhances accuracy, reduces administrative workload, and improves overall efficiency in classroom management.

Keywords—Attendance System, Virtual Card, Near Field Communication (NFC), Host Card Emulation, Mobile Application.

I. INTRODUCTION

Attendance monitoring is an essential procedure in higher education institutions because it is frequently used to assess student engagement, uphold academic rules, and assist with course administration. Nevertheless, traditional documenting attendance, such as physical roll calls, manual sign-in sheets, and simple digital forms are ineffective and vulnerable to fake attendance and human mistakes. These drawbacks become more noticeable in large classes because monitoring attendance might interfere with instruction and lower productivity.

As smartphones become more widely used, mobile solutions have arisen as effective options for automating administrative duties. Near Field Communication (NFC) technology allows for secure data transfer over short distances between devices and is extensively used in fields like access control and digital identification. With Host Card Emulation (HCE), mobile devices are able to act as virtual cards without the need for physical credentials. Utilizing these technologies, this project presents a Virtual Card Attendance System that enables students to mark their attendance with their smartphones. The system strives to enhance precision, lessen administrative burden, and offer a

reliable and effective attendance management system appropriate for university settings.

II. METHODOLOGY

This section outlines the approach used in the creation of the Virtual Card Attendance System. The methodology offers an organized framework to direct the project from requirements analysis to system implementation and assessment. In that case, it guarantees that every stage of development is methodically planned, carried out and evaluated to produce a dependable and user-friendly solution. The chosen methodology places a strong emphasis on early prototyping, iterative development and continuous improvement, which are especially appropriate for the creation of mobile applications. The important tasks of this methodology are covered, such as the requirement specification, system architecture design, feature implementation and functional testing. The development process stays flexible and well-organized, allowing for the prompt identification of problems, the mitigation of development risks and the alignment of the finished system with project goals and user requirements by using this structured approach.

A. Development Approach

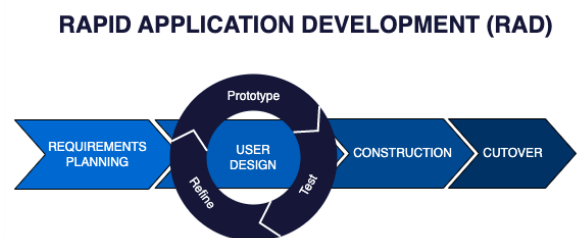


Figure 1: Rapid Application Development

In this project, we implement the Rapid Application Development (RAD) approach to direct the creation of the Virtual Card Attendance System because RAD emphasizes quick prototyping, iterative development and ongoing user involvement, which are ideal for mobile application systems

with changing requirements. The four primary stages of the techniques are requirements planning, user design, quick building and cutover. System goals and functional requirements are determined based on current attendance procedures during the requirements planning stage. Through early prototypes, the user design phase concentrates on interface layout and workflow design, enabling adjustments depending on feedback.

B. Technical Implementation

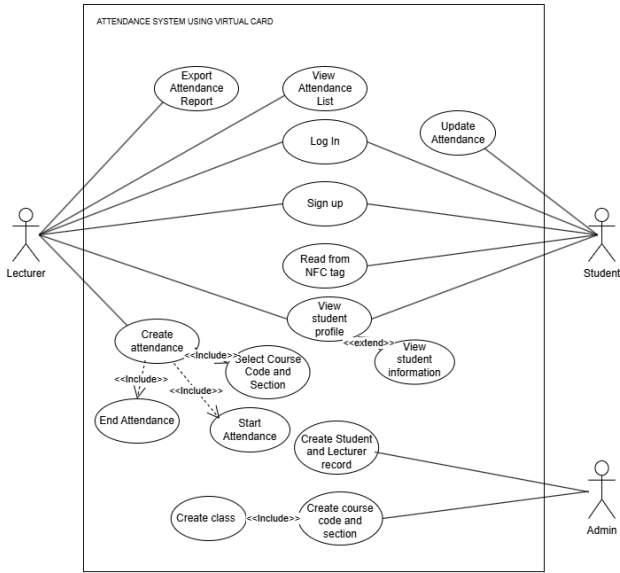


Figure 2: Use Case Diagram

As shown in Figure 2.2, the technological implementation of the Attendance System Using Virtual Card is intended to facilitate role-based interactions between instructors, students, and administrators. The Flutter framework is used to develop the system as a mobile application, which is coupled with a cloud-based backend to guarantee real-time data processing and synchronization. User identification, course and class management, attendance session control, and NFC-based attendance recording are examples of core features.

Students engage with the system by creating accounts, logging in, and marking attendance via NFC communication, while professors are in charge of setting up and overseeing attendance sessions, examining attendance lists and exporting attendance reports. Administrative tasks include maintaining user records, class information and course codes to ensure data consistency throughout the system. In order to provide precise and centralized record maintenance, attendance data obtained through NFC interactions is sent to the backend for validation and storage. This implementation strategy guarantees effective attendance-related operations in a university setting, permits smooth communication between system components, and offers secure role-based access.

I. Frontend Architecture

The main layer of user-system interaction is the frontend architecture of the Attendance System Using Virtual Card. The Flutter framework, which offers a unified and consistent user interface across compatible devices, is used to construct the mobile application. Attendance session interfaces, authentication screens and attendance history displays are just a few of the user-facing elements that Flutter controls. User inputs and system actions are handled locally before being converted into structured data and safely sent to the backend services. In order to guarantee responsive interaction, a clear navigation flow and effective handling of user activities during attendance marking, the frontend architecture was created. The system provides a dependable and seamless user experience while facilitating easy contact with the backend infrastructure by upholding a well-organized display layer.

II. Backend Architecture

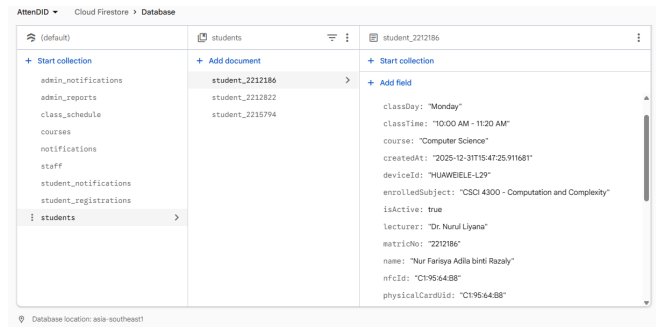


Figure 3: Collection of student_registration

Firebase is a cloud-based platform used in the implementation of the Attendance System Using Virtual Cards' backend architecture to facilitate data administration, real-time synchronization and authentication. Firebase Authentication is utilized to confirm user identities and control role-based access for administrators, instructors and students. Firebase Firestore is used to store attendance-related data. Student_registration kept these structured records in a collection such as student identification details, course information, device identifiers, registration status and activation timestamps. In order to facilitate effective retrieval and validation during attendance sessions, each document in the collection represents a distinct student record. The backend design facilitates real-time updates between mobile devices and the database while guaranteeing data access through restricted permissions. The system's scalability, data integrity and dependable backend operations for tracking attendance in a university setting are all made possible by this design.

III. Database Architecture

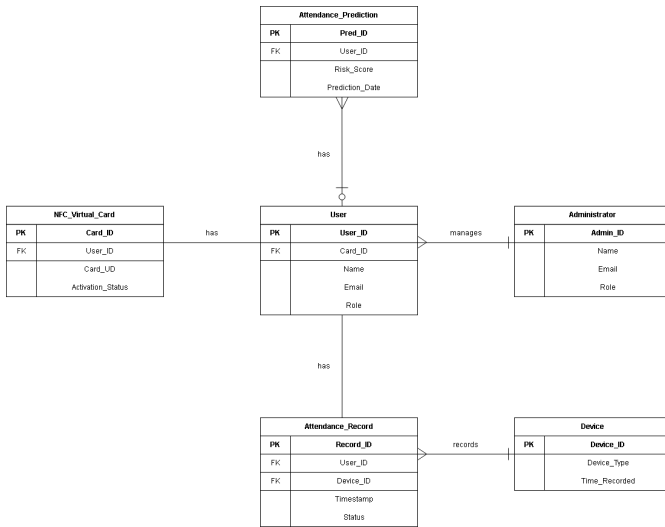


Figure 4: Database Design

Firestore is used in the database architecture of the Attendance System Using Virtual Card to enable organized, safe and effective storage of attendance-related data. Users, virtual cards, attendance records, gadgets and administrative management are among the many interconnected entities that make up the database. Every user is uniquely identifiable and linked to a virtual NFC card that holds card-identifying details and activation status. The system may capture attendance status and timestamps for verification because attendance data are connected to both user and device entities. Administrative organisations oversee system setup and user roles, guaranteeing restricted access to private information. Accurate attendance activity tracking is made possible by this relational structure, which also preserves data scalability and integrity. The database design provides instantaneous data updates across devices by utilising Firestore's real-time synchronisation capabilities, guaranteeing dependable and consistent attendance records in the academic setting.

III. RESULTS

A. User Interface

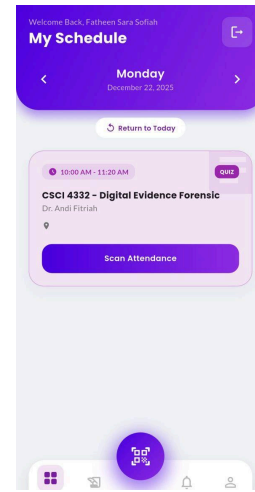


Figure 5: Student Schedule Interface

Figure 5 shows the Student Schedule Dashboard interface, which is essentially the first page of the student application interface. The interface has a horizontal date selection tool at the top of the page for navigation of the weekly schedule of activities for the week. In the middle of the interface is the current ongoing class in terms of the subject title "CSCI 4332 - Digital Evidence Forensic," as well as other details including the time slot in which the lesson takes "10:00 AM - 11:20 AM," the location of the lesson, and the identity of the teacher of the subject matter. Most importantly, this interface has the "Scan Attendance" button integrated into it.

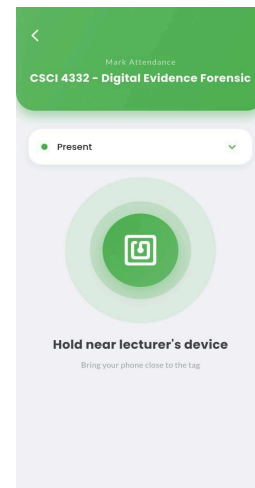


Figure 6: Active Scanning Interface

Figure 6 displays the Active Scanning Interface, which appears once the student initiates the attendance process. The UI is displayed once the student decides to initiate the attendance-taking process. The UI is shown with an animation centered around the NFC icon, emphasizing that the NFC is functional. The UI contains the directive prompt "Hold near lecturer's device" that enables the student to

perform the action without errors. The specific context for the current course "CSCI 4332" is shown at the top of the header.

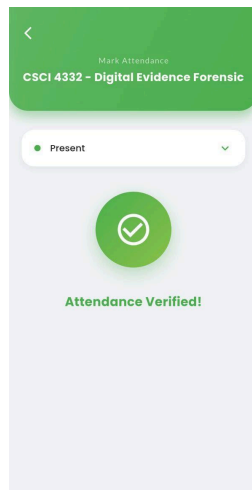


Figure 7: Attendance Verification Interface

Figure 7 above shows the Verification Confirmation Interface, which appears immediately after a correct NFC handshake has occurred. The interface features a prominent tick mark icon coupled with an "Attendance Verified!" notification, acting as an instant trigger of positive reinforcement. The status bar clearly indicates "Present," attesting to the indication of an immediate update in the back-end database. The unique visual cue, from the pulsating effect to a final success state, concludes the loop of user feedback, enabling a safe exit from the application.

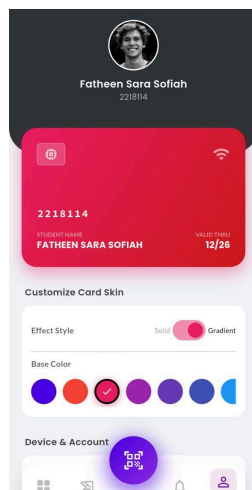


Figure 8: Student Profile Interface

Figure 8 shows the Student Profile Interface, representing the digital identity of the user on the platform.

On top, it displays the static details of the student himself, such as his profile picture, full name, and Matric ID. In the middle of this page is the Virtual Card, situated at the center with a background image of the actual student ID card, with the due dates and the name of the student who is in possession. The skin customization panel below the card displays an interface where users can input their favorite gradient and base color for their digital ID, the skin of the ID, which is one of the added features to keep users engaging with the application.

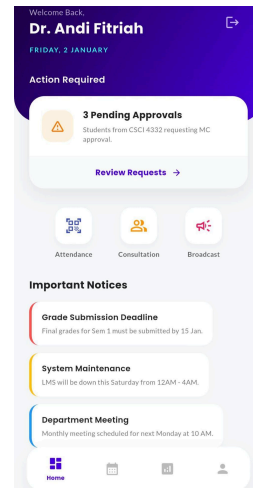


Figure 9: Staff Dashboard Interface

Figure 9 shows a display of Staff Dashboard Interface with a functionality designed to act as a command center for lecturers. The dashboard interface welcomes its user with a personalized greeting and highlights messages requiring action tasks, such as requests for approvals. The functionality and services required are provided through three separate quick action buttons – "Attendance" for conducting class viewing sessions, "Consultation" for handling student consultation sessions, and "Broadcast" for broadcasting messages to all users. The section below the navigation bar contains an "Important Notices" area where important system notices are summarized to keep staff members well-informed on system requirements.

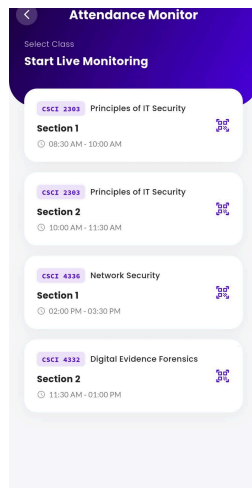


Figure 10: Attendance Monitor Interface

In Figure 10 is the Attendance Monitor Selection Interface. Here, staff members are able to choose the particular class session to monitor attendance. In this case, the attendance is categorized based on the course number and section number. Additionally, the time the classes are on is clearly shown. Every card on the list is marked with its own particular icon on the right. This is to show the employee that the particular card has its own scanning module to initiate the particular class session.

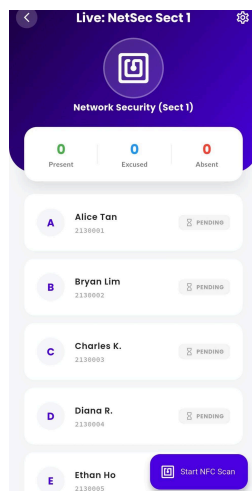


Figure 11: Live Attendance Monitoring Interface

Figure 11 depicts the Live Attendance Monitoring Interface, which is the main operational interface during the class session. The header contains the description of the running session, as well as the representation of the presence statistics: Present, Excused, and Absent. The body contains the list of enrolled students, displayed alongside their status represented as "Pending" by default. There is a floating action button at the bottom right corner labeled "Start NFC Scan" that, when toggled, enables the device's

NFC scanner to start receiving student cards. This screen is refreshed in real-time as cards are scanned.

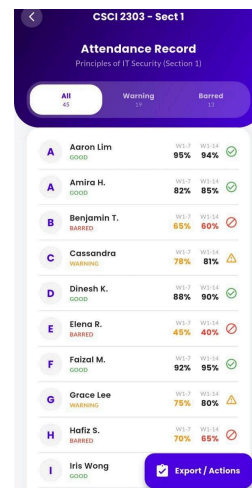


Figure 12: Attendance Record Interface

Figure 12 presents the Attendance Record Interface, which serves as the analytical component of the staff module. This visualization aids lecturers in quickly assessing class engagement levels. Furthermore, the report screen provides options to finalize the attendance record or release warning letters to students who have fallen below the required attendance threshold, integrating administrative enforcement directly into the workflow.

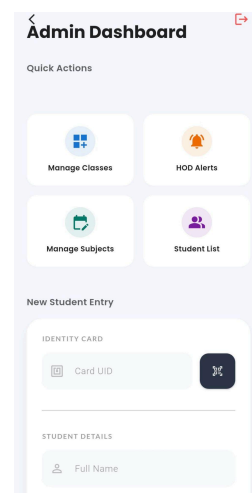


Figure 13: Admin Dashboard Interface

In Figure 13, it is shown how the Admin Dashboard Interface looks and functions as the command center for system administrators. This interface is designed for rapid navigation and includes a "Quick Actions" grid that enables quick selection of the four main modules, which include Manage Classes, HOD Alerts, Manage Subjects, and the Students List. At the bottom of the "Quick Actions" grid is

placed the New Students' Entry form for easy use and process optimization. This form includes a scan button for a "Card UID" for easy and direct linking to a student profile without necessarily leaving the main dashboard.

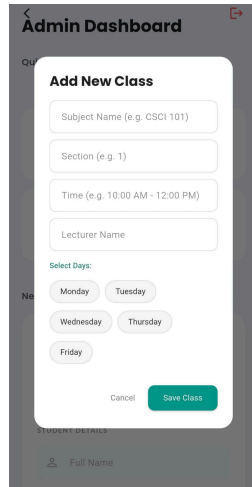


Figure 14: Class Creation Interface

In Figure 14 is the "Class Creation Interface," used for entering new offerings of classes into the master schedule of the system. In this interface is the essential data for each class session: the Subject Name, Section Number, Time Duration of the lecture, along with connecting it to the particular Lecturer for the class so that the right individual has control over the attendance monitoring process for this particular class session of the subject. This interface also supports the selection of days for the offering of the particular subject for more precise scheduling of classes in the master schedule of the particular learning environment in the system.

Figure 15 displays the Student Database Interface, which provides the overall student details registered in the database. The header contains the overall statistics, indicating the total, active, and unclaimed students. There is the search feature that enables the admins to search the database based on the name or ID. The filters are used for the overall categories. Individually, each student is displayed in the form of a card that is detailed with the student name, Matric ID, as well as the current enrollment course. Importantly, each card contains individual administrative operations such that students can be edited for changes, "Unbind" for the lost or dysfunctional NFC card, along with the "Delete" option that will delete the account altogether.

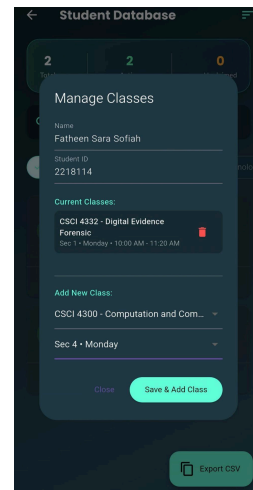


Figure 16: Class Enrollment Interface

Figure 16 shows the Class Enrollment Interface. This is an interface accessed through a pop-up feature when an admin clicks on the "Edit" and "Manage Classes" button corresponding to a particular student. This interface displays the "Current Classes" enrolled by the particular student and allows deletion of particular classes through deletion buttons represented by trash symbols. The "Add New Class" sub-module provides subject and time slot menus for new class addition. This is crucial for updating the database on the student's class timetable, fundamental for valid entry requirements during attendance scanning.

B. Evaluation

The interfaces developed and their testing have demonstrated that the system integrates rapid NFC-based scanning protocols with centralized administrative control, ensuring seamless real-time data synchronization along with operational efficiency across the Student, Staff, and Admin modules.

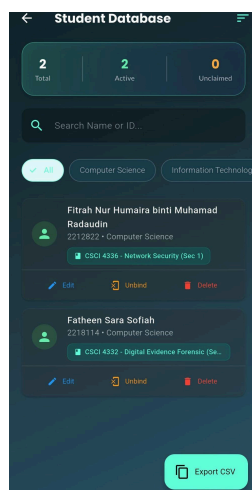


Figure 15: Student Database Interface

IV. DISCUSSION

The development of AttenDID demonstrates the feasibility of replacing manual attendance tracking with a secure, NFC-based mobile solution. The system successfully achieved its primary objective of converting physical card data into a secure digital identity, enabling automated attendance recording without the need for specialized biometric hardware.

A critical contribution of this study is the mitigation of “buddy punching” through device binding. By linking a student's account strictly to a single smartphone, the system effectively prevents students from sharing login credentials to mark attendance for absent peers. Furthermore, the integrity of the attendance data is enforced by the Admin pre-registration, which ensures that only physical cards explicitly authorized by the administration can activate a virtual profile, automatically rejecting unregistered or cloned tags.

In terms of non-functional requirements, the system met the performance benchmark of processing an NFC scan and updating the database within two seconds. This rapid processing speed addresses the bottlenecks associated with paper-based sign-in sheets. The usability testing results further validate the design, with 72.5% of respondents expressing readiness to use the system regularly and over 70% agreeing that it is faster than traditional methods.

The implementation of a Client-Server architecture using Flutter and Firebase proved effective for real-time applications. The NoSQL structure of Cloud Firestore allowed for scalable read/write operations, ensuring that the system remained stable even if multiple attendance records were generated simultaneously during class entry.

A. Limitations

The system's deployment is currently constrained by its strict dependency on smartphones equipped with embedded NFC hardware, which limits accessibility for users with older or budget devices that lack active NFC sensors. Additionally, the Host Card Emulation (HCE) module faces stability challenges on specific Android devices where aggressive battery optimization and proprietary security features can disrupt background NFC services, rendering the “Tap-to-Attend” function unstable. Furthermore, the solution lacks cross-platform compatibility, as iOS implementation remains untested due to the restrictive nature of Apple's core NFC framework and associated developer licensing requirements.

B. Recommendations For Future Works

To address current limitations and improve system robustness, future development will focus on three key areas. First, biometric authentication, such as Face ID or fingerprint scanning, will be integrated as a secondary validation layer before the NFC scan to ensure the user is the registered student. Second, a local database architecture will be implemented to enable offline attendance marking, allowing data to be stored locally and synchronized with the cloud once internet connectivity is restored. Finally, the system will be optimized for cross-platform compatibility, specifically targeting iOS devices to ensure accessibility for all students regardless of their smartphone ecosystem.

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